

Lecture 7

§1 Likelihood

1. Definition: Likelihood, log-likelihood, likelihood estimator

令 $X = \{X_1, \dots, X_n\}$ 有 distribution function $X \sim f(x|\theta)$, $\theta \in \Theta$, 则

① likelihood function 被定义为

$$L(\theta|X) = f(X|\theta)$$

② log-likelihood function 被定义为

$$l(\theta) = \ln L(\theta|X) = \log L(\theta) = \log L(\theta|X)$$

③ maximum likelihood estimator (MLE) 被定义为

$$\hat{\theta} = \operatorname{argmax}_{\theta \in \Theta} L(\theta|X) = \operatorname{argmax}_{\theta \in \Theta} l(\theta|X)$$

注: MLE 不一定存在, 例如: $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} \text{Unif}(0, \theta)$, 则

$$L(\theta|X) = \frac{1}{\theta^n} \cdot 1_{\{\max_{1 \leq i \leq n} X_i \leq \theta\}}$$

注意到 $\theta \in \Theta = (X_{(n)}, \infty)$, 因此 θ 实际上取不到 $X_{(n)}$.

为了避免这种情况, 可以将 MLE 定义为:

$$\hat{\theta} = \operatorname{argmax}_{\theta \in \Theta} L(\theta|X) = \operatorname{argmax}_{\theta \in \Theta} l(\theta|X) \quad (\theta \text{ 在 } \Theta \text{ 的 closure 上取值})$$

接下来, 假设 $X = \{X_1, \dots, X_n\}$ 服从 $f(x|\theta_0) \in \mathcal{F} = \{f(x|\theta), \theta \in \Theta\}$, 即 θ_0 为真实值, 将 $\hat{\theta}_{MLE}$ 记作 θ_0 的 MLE

2. Definition: Consistency condition of MLE (MLE 的 consistency 的条件)

① Θ 为 compact, 或

Θ 满足 $\sup_{\theta \in \Theta, |\theta - \theta_0| \geq \varepsilon} E[l(\theta)] < E[l(\theta_0)]$, $\forall \varepsilon > 0$ (separation condition)
($E[l(\theta_0)]$ 严格大于其他 $E[l(\theta)]$ 的 supremum)

② $\frac{1}{n} l(\theta)$ converges uniformly to $\frac{1}{n} E[l(\theta)]$ in probability, 即

$$\sup_{\theta \in \Theta} \frac{1}{n} |l(\theta) - E[l(\theta)]| \xrightarrow{P} 0$$

3. Theorem: Consistency of MLE

若 consistency condition of MLE 满足,

则 $\hat{\theta}_{MLE}$ 为 θ_0 的一个 consistent estimator, 即 $\hat{\theta}_{MLE} \xrightarrow{P} \theta_0$

* 证明:

Proof of Theorem. 1.3. Notice that,

$$\hat{\theta}_{MLE} = \operatorname{argmax}_{\theta \in \Theta} \ell(\theta) = \operatorname{argmax}_{\theta \in \Theta} \left[\frac{1}{n} \sum_{i=1}^n \log \frac{f(X_i|\theta)}{f(X_i|\theta_0)} \right] \triangleq \operatorname{argmax}_{\theta \in \Theta} M(\theta)$$

Since we have

$$M(\theta) = \frac{1}{n} \sum_{i=1}^n \log \frac{f(X_i|\theta)}{f(X_i|\theta_0)} \rightarrow \mathbb{E} \left[\log \frac{f(X_i|\theta)}{f(X_i|\theta_0)} \right] = -KL(f_{\theta_0} \| f_{\theta}) < 0,$$

where the last inequality sign changes to the equal sign iff $f(x|\theta) \equiv f(x|\theta_0)$, a.s., i.e., $\theta = \theta_0$. Therefore, if Θ is compact, then for $\forall \varepsilon > 0$, we have

$$\sup_{\theta \in \Theta, |\theta - \theta_0| \geq \varepsilon} \mathbb{E} M(\theta) < \mathbb{E} M(\theta_0) = -KL(f_{\theta_0} \| f_{\theta_0}) = 0. \quad (3.1)$$

Similarly we can also conclude (3.1) under the separation condition (1.1). Now, if we denote

$$\delta = \mathbb{E}M(\theta_0) - \sup_{\theta \in \Theta, |\theta - \theta_0| \geq \epsilon} \mathbb{E}M(\theta) > 0,$$

Since $M(\theta)$ converges uniformly to $\mathbb{E}M(\theta)$ in probability according to (ii) of Condition 1.2. Therefore, there exists $N \in \mathbb{N}^+$ s.t. when $n \geq N$,

$$\begin{aligned} \mathbb{P}(|\hat{\theta}_{MLE} - \theta_0| \geq \epsilon) &= \mathbb{P}\left(\sup_{\theta \in \Theta, |\theta - \theta_0| \geq \epsilon} M(\theta) > M(\theta_0)\right) \\ &\leq \mathbb{P}\left(\sup_{\theta \in \Theta, |\theta - \theta_0| \geq \epsilon} [M(\theta) - \mathbb{E}M(\theta)] > [M(\theta_0) - \mathbb{E}M(\theta_0)] + \delta\right) \\ &\leq \mathbb{P}\left(\sup_{\theta \in \Theta, |\theta - \theta_0| \geq \epsilon} [M(\theta) - \mathbb{E}M(\theta)] > \frac{\delta}{2}\right) + \mathbb{P}\left([M(\theta_0) - \mathbb{E}M(\theta_0)] > -\frac{\delta}{2}\right) \\ &\leq 2\mathbb{P}\left(\sup_{\theta \in \Theta} |M(\theta) - \mathbb{E}M(\theta)| > \frac{\delta}{2}\right) \rightarrow 0. \end{aligned}$$

Hence $\hat{\theta}_{MLE} \xrightarrow{P} \theta_0$. □

4. Definition: Consistency and Asymptotic Normality (CAN) condition of MLE

对于 $\Theta \in \Theta$, 满足

- ① θ_0 是 Θ 的一个 interior
- ② $f(x|\theta)$ 有一个不取决于 θ 的 common support $\mathcal{X}$
- ③ partial derivative 和 integral 可交换, 即

$$\frac{\partial^i}{\partial \theta^i} \mathbb{E}[\log f(x|\theta)] = \mathbb{E}\left[\frac{\partial^i}{\partial \theta^i} \log f(x|\theta)\right], \text{ for } i=1,2$$

- ④ score function 的 derivative $s'(\theta|x)$ (即 $l''(\theta)$) 关于 θ 连续
- ⑤ $0 < I(\theta) < \infty$ 为 well-defined

5. Theorem: Consistency and Asymptotic Normality of MLE

若 ① consistency condition of MLE 满足,

② CAN condition of MLE 满足,

则 $\sqrt{n}(\hat{\theta}_{MLE} - \theta_0) \xrightarrow{d} N(0, I(\theta_0)^{-1})$ (这里的 $I(\theta_0)$ 是 sample size 为 1 的时候的)

证明:

• 若 θ_0 是 Θ 的一个 interior, 且

$\hat{\theta}_{MLE}$ 为 θ_0 的 consistent estimator,

则 $\hat{\theta}_{MLE}$ 也是 Θ 的一个 interior (n 足够大时)

$$\Rightarrow l'(\hat{\theta}_{MLE}) = 0$$

• 由 Taylor's expansion, 有

$$0 = l'(\hat{\theta}_{MLE}) = l'(\theta_0) + l''(\hat{\theta})(\theta_0 - \hat{\theta}_{MLE}) \quad (\text{关键步骤, 在 Delta method 中出现过})$$

$$\Rightarrow \sqrt{n}(\hat{\theta}_{MLE} - \theta_0) = \sqrt{n} \cdot \frac{l'(\theta_0)}{l''(\hat{\theta})} = \frac{\frac{1}{\sqrt{n}} l'(\theta_0)}{\frac{1}{n} l''(\hat{\theta})}, \text{ 接下来分别研究分子分母的渐近分布即可}$$

其中 $\hat{\theta}$ lies in between θ_0 和 $\hat{\theta}_{MLE}$ ($\hat{\theta} \xrightarrow{P} \theta_0$)

- 由 CAN condition 的 ③ 和 CLT, 有

$$\frac{1}{\sqrt{n}} \ell'(\theta_0) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial \log f(X_i | \theta)}{\partial \theta} \Big|_{\theta=\theta_0} \xrightarrow{d} N(E[S(\theta_0 | X_i)], I(\theta_0)) \stackrel{d}{=} N(0, I(\theta_0))$$

交换微分和积分

- 由 CAN condition 的 ③ 和 WLLN, 有

$$\frac{1}{n} \ell''(\theta_0) = \frac{1}{n} \sum_{i=1}^n \frac{\partial^2 \log f(X_i | \theta)}{\partial \theta^2} \Big|_{\theta=\theta_0} \xrightarrow{P} E\left[\frac{\partial^2 \log f(X_i | \theta)}{\partial \theta^2}\right] \Big|_{\theta=\theta_0} = -I(\theta_0)$$

交换微分和积分

- 综上, 由 Slutsky's theorem, 有

$$\sqrt{n}(\hat{\theta}_{MLE} - \theta_0) = \frac{\frac{1}{\sqrt{n}} \ell'(\theta_0)}{\frac{1}{n} \ell''(\theta_0)} \xrightarrow{d} N(0, I(\theta_0)^{-1})$$

b. Definition: $g(\theta)$ 的 log-likelihood

若 θ 的 log-likelihood function 为 $\ell(\theta)$,

则对于任意 function $g(\theta)$, $g(\theta)$ 的 log-likelihood function 被定义为:

$$\ell_g(\theta) \triangleq \max_{\theta \in \Theta, g(\theta) = \theta} \ell(\theta)$$

$g(\theta) = \theta$ 时的 log-likelihood = 先限定 θ 满足 $g(\theta) = \theta$, 再 maximize $\ell(\theta)$

注: 这么定义是为了满足 Invariance property of MLE

7. Property: Invariance property of MLE

若 θ 的 MLE 为 $\hat{\theta}_{MLE}$,

则对任意 function $g(\theta)$, $g(\theta)$ 的 MLE 为 $g(\hat{\theta}_{MLE})$

注: idea: 令 $\eta = g(\theta)$, 则

$$\hat{\eta} = \arg\max_{\eta} \ell_g(\eta) \quad (\text{MLE 的定义})$$

$$= \arg\max_{\eta} \max_{\theta \in \Theta, g(\theta) = \eta} \ell(\theta)$$

$$\leq \ell(\hat{\theta}_{MLE} | x), \text{ 当 } \eta = g(\hat{\theta}_{MLE}) \text{ 时取 "="}$$

换言之, 对于 $\forall \eta$, $\ell(\eta) \leq \ell(\hat{\theta}_{MLE} | x)$ 一定成立.

但是根据 $g(\theta)$ 的 log-likelihood function 的定义, $\eta = g(\hat{\theta}_{MLE})$ 时能取等.

因此 $\hat{\eta}_{MLE} = g(\hat{\theta}_{MLE})$

证明:

Proof of Theorem. 1.7. We proof by contradiction. If there exist some other $\theta_0 \neq \hat{\theta}_{MLE}$ such that $\phi_0 = g(\theta_0) \neq g(\hat{\theta}_{MLE})$, and $\phi_0 = g(\theta_0)$ is the MLE of $g(\theta)$. Then

$$\ell_g(\phi_0) = \max_{\theta \in \Theta, g(\theta) = \phi_0} \ell(\theta) > \ell_g(g(\hat{\theta}_{MLE})) = \max_{\theta \in \Theta, g(\theta) = g(\hat{\theta}_{MLE})} \ell(\theta) = \ell(\hat{\theta}_{MLE}),$$

which contradict with the fact that $\hat{\theta}_{MLE}$ being the MLE of $\ell(\theta)$. Hence $g(\hat{\theta}_{MLE})$ is the MLE of $g(\theta)$. $\square$

例 1: • Example 1.9 (♣ MLE in Multinomial Distribution). Suppose there are k scenarios each with probability p_i to occur respectively, $1 \leq i \leq k$, i.e.

$A_1, \dots, A_k$ are disjoint with $\Omega = \bigcup_{i=1}^k A_i$, and $\mathbb{P}(A_i) = p_i$, for $1 \leq i \leq k$,

Apparently, $\sum_{i=1}^k p_i = 1$. We repeat the trial independently for n times. Each time j , $1 \leq j \leq n$, we observe one of the scenarios happens and record $Y_j = (Y_{j1}, \dots, Y_{jk}) = (0, \dots, 0, 1, 0, \dots, 0)_{k \times 1}$ with s -th element being 1 if s -th element occurs. We denote X_i as the number of times A_i occurs, so we have the sample $X = (X_1, \dots, X_k) = (x_1, \dots, x_k)$ and knowing that $X = (X_1, \dots, X_k)$ follows a multinomial distribution. Please find the MLE for $p_1, \dots, p_k$ and please derive the CAN property of the MLE.

• 由于 $X = (X_1, \dots, X_k) \sim \text{Multinomial}(n, p_1, \dots, p_k)$ 且 $\sum_{i=1}^k p_i = 1$, 则

$$L(p, X) = \frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k} \cdot 1\{0 \leq p_i \leq 1, \sum_{i=1}^k p_i = 1\}$$

$$\Rightarrow l(p, X) = \log(n!) - \sum_{i=1}^k \log x_i! + \sum_{i=1}^{k-1} x_i \log p_i + x_k \log(1 - \sum_{i=1}^{k-1} p_i)$$

$$\frac{\partial}{\partial p_i} l(p, X) = 0$$

$$\Rightarrow \begin{cases} \frac{x_i}{p_i} - \frac{x_k}{1 - \sum_{i=1}^{k-1} p_i} = 0 & \text{for } i = 1, \dots, k-1 \\ \sum_{i=1}^k x_i = n \end{cases}$$

$$\text{解得: } \hat{p}_{i, \text{MLE}} = \frac{x_i}{n}$$

• 由于 $Y_j = (Y_{j1}, \dots, Y_{jk}) = (0, \dots, 0, \underbrace{1}_{\text{第 } s \text{ 个}}, 0, \dots, 0)$ for $j = 1, \dots, n$, 有

$$X = \sum_{j=1}^n Y_j$$

$$\Rightarrow X_i = \sum_{j=1}^n Y_{ij}$$

由于 $Y_{ij} \sim \text{Ber}(p_i)$, 有 $\text{var}(\frac{x_i}{n}) = \frac{1}{n} p_i (1 - p_i)$, 因此

$$\sqrt{n}(\hat{p}_{i, \text{MLE}} - p_i) \xrightarrow{d} N(0, p_i(1 - p_i))$$

例 2: (MLE 不一定唯一) 令 $X = \{X_1, \dots, X_n\}$ 为 random sample from $\text{Unif}(\theta, \theta+1)$, 求 θ 的 MLE

$$f(x|\theta) = 1\{X_{(1)} \geq \theta, X_{(n)} \leq \theta+1\}$$

$$\Rightarrow \hat{\theta}_{\text{MLE}} \in [X_{(n)} - 1, X_{(1)}]$$

例 3: • Example 1.11 (♣ MLE in Generalized Linear Models). Assume the sample $X = \{X_1, \dots, X_n\}$ has mutually independent observations $X_1, \dots, X_n$. Each X_i has probability distribution function

$$f_{X_i}(x_i|\eta_i, \phi_i) = \exp\left(\frac{\eta_i x_i - \zeta(\eta_i)}{\phi_i}\right) \cdot h(x_i, \phi_i), \quad \text{使用 canonical link}$$

where $\{\phi_i > 0, 1 \leq i \leq n\}$ is called the dispersion parameters and here we assume there are known for simplicity. Apparently, we have f_{X_i} forms an exponential family and by the differentiation of exponential family density, we have

$$0 = \frac{\partial}{\partial \eta_i} \int f_{X_i}(x_i|\eta_i) dx_i = \int \frac{\partial f_{X_i}(x_i|\eta_i)}{\partial \eta_i} dx_i = \frac{1}{\phi_i} [\mathbb{E}X_i - \zeta'(\eta_i)] = \frac{1}{\phi_i} [x_i - \zeta'(\eta_i)] \cdot f(x_i|\eta_i, \phi_i)$$

Hence $\mathbb{E}X_i = \zeta'(\eta_i)$ for $1 \leq i \leq n$. It's assumed that the mean can be explained by known covariates (effects) Z_i and interested effect size β as

$$g(\mathbb{E}X_i) = g(\zeta'(\eta_i)) = \beta^T Z_i, \quad i = 1, \dots, n$$

for some known link function g , here we consider $g \circ \zeta'$ to be the identical function for simplicity, i.e., $\eta_i = \beta^T Z_i$. If ζ is smooth and convex (i.e., $\zeta''(x) > 0$ for all x in its domain), please give the MLE of β .

log-likelihood function 为

$$l(\beta) = \sum_{i=1}^n \left[\log h(x_i, \theta_i) + \frac{\beta^T z_i x_i - \xi(\beta^T z_i)}{\theta_i} \right]$$

对 β 求导, 有

$$\frac{\partial l(\beta)}{\partial \beta} = \sum_{i=1}^n \frac{1}{\theta_i} (x_i - \xi'(\beta^T z_i)) z_i$$

$$\frac{\partial^2 l(\beta)}{\partial \beta^2} = - \sum_{i=1}^n \frac{1}{\theta_i} \xi''(\beta^T z_i) \cdot z_i z_i^T < 0$$

因此 MLE β 满足:

$$\sum_{i=1}^n \frac{1}{\theta_i} (x_i - \xi'(\hat{\beta}^T z_i)) z_i = 0$$

注: 在 STA4102 中, MLE β 满足 $U(\beta) = \frac{1}{\sigma^2} \sum_{i=1}^n (y_i - \mu_i)' X_i = 0$, 结果一致